

Graph of Drafts

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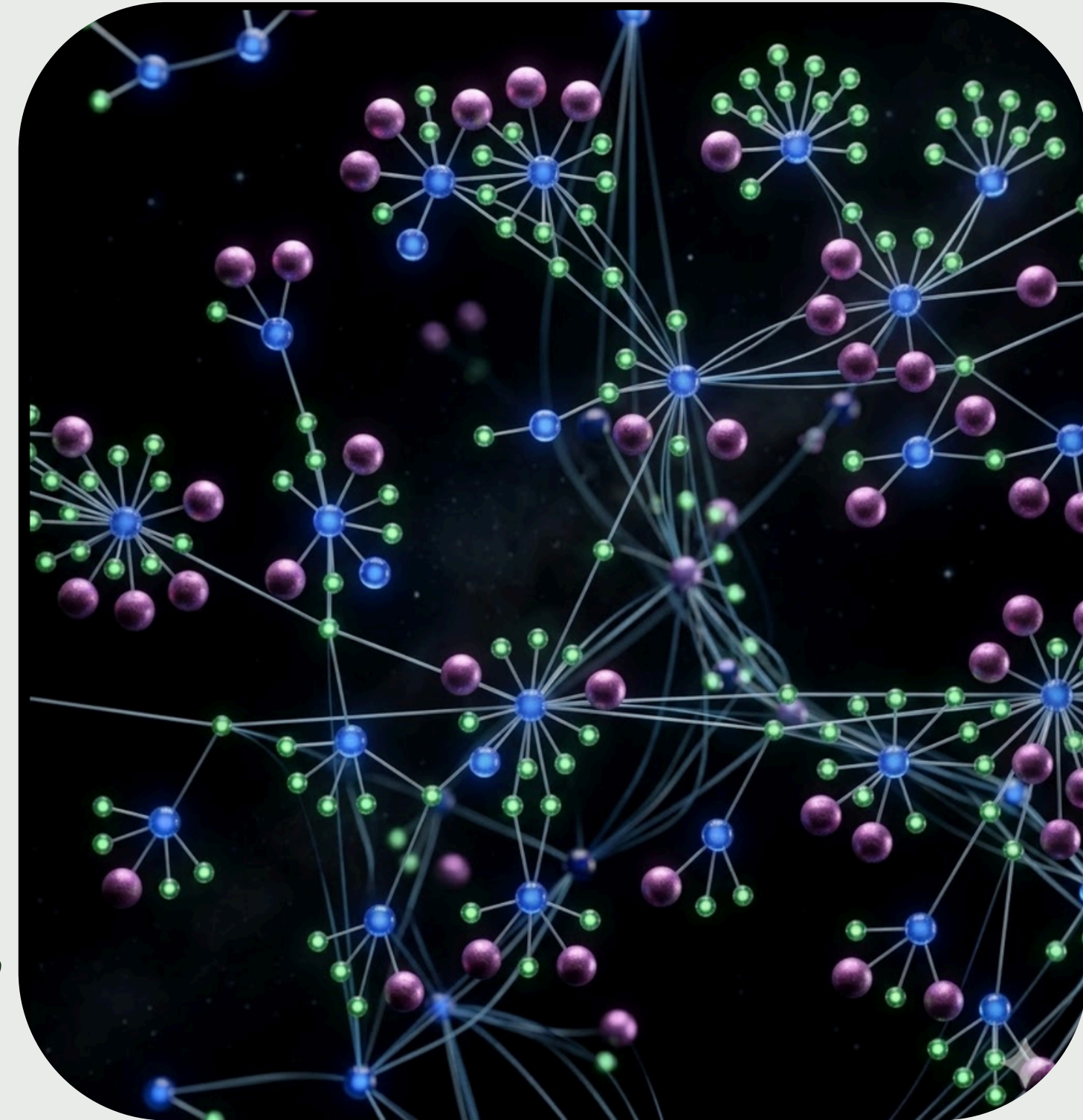
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Table of Contents

- Introduction
- Related Works
- Limitations
- Graph of Drafts
- Operational Flow
- System Architecture
- Evaluations - Setup & Usecases
- Results and Discussions
- Conclusions
- References
- Appendix

Introduction

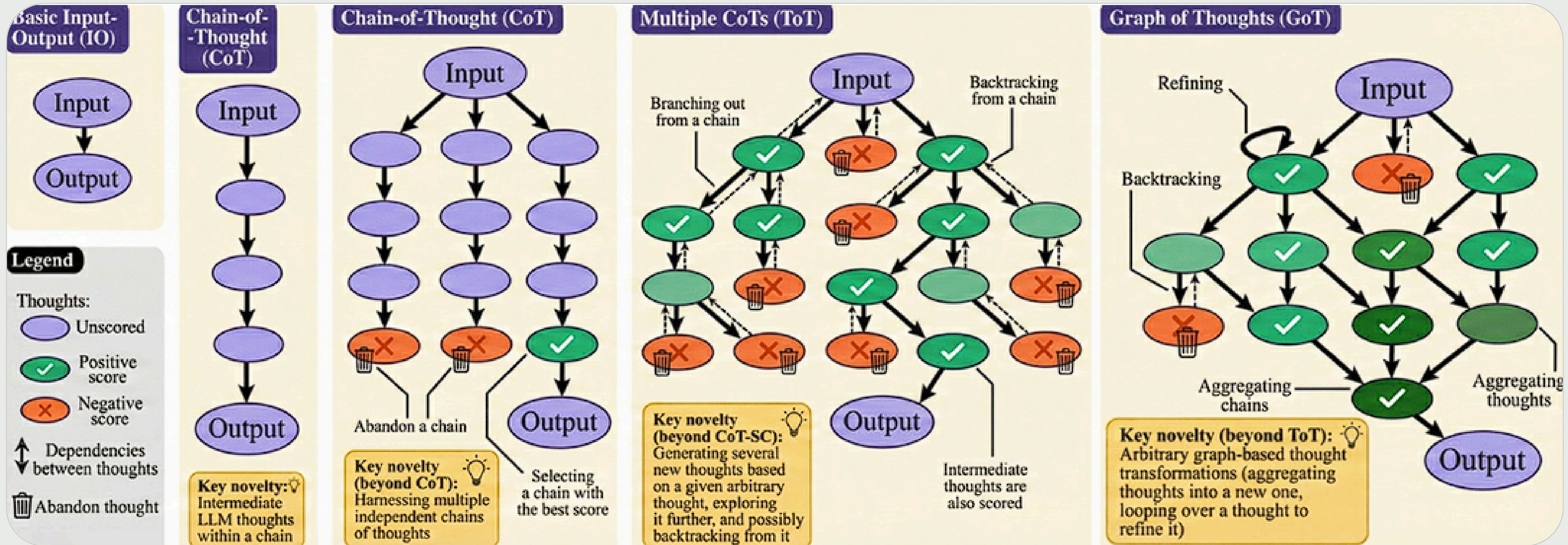
- LLM agents need reasoning frameworks that mirror how humans actually think.
- Graph of Thoughts advanced beyond contemporary frameworks, including CoT, CoD, and ToT, by introducing graph-based reasoning, yet most of these faces verbosity and structural disadvantages.
- Graph of Drafts introduces bidirected graphs built from concise drafts, enabling peer synergy, in-place refinement, and contextual richness with significantly less token overhead



Related Works

- ***Chain-of-Thought***: Sequential step-by-step reasoning embedded in prompts, constrained to a single linear chain with no branching or backtracking and token overhead.
- ***Chain-of-Draft***: Minimalist variant of CoT that substitutes verbose reasoning steps with concise information-dense drafts, reducing token overhead while retaining single-pass structure.
- ***Tree-of-Thought***: Hierarchical branching over multiple reasoning paths with backtracking support, though sibling branches remain structurally isolated with no cross-branch information sharing.
- ***Graph-of-Thought***: Directed graph representation of reasoning enabling thought merging and aggregation, but unidirectional edges preclude peer-level interaction.

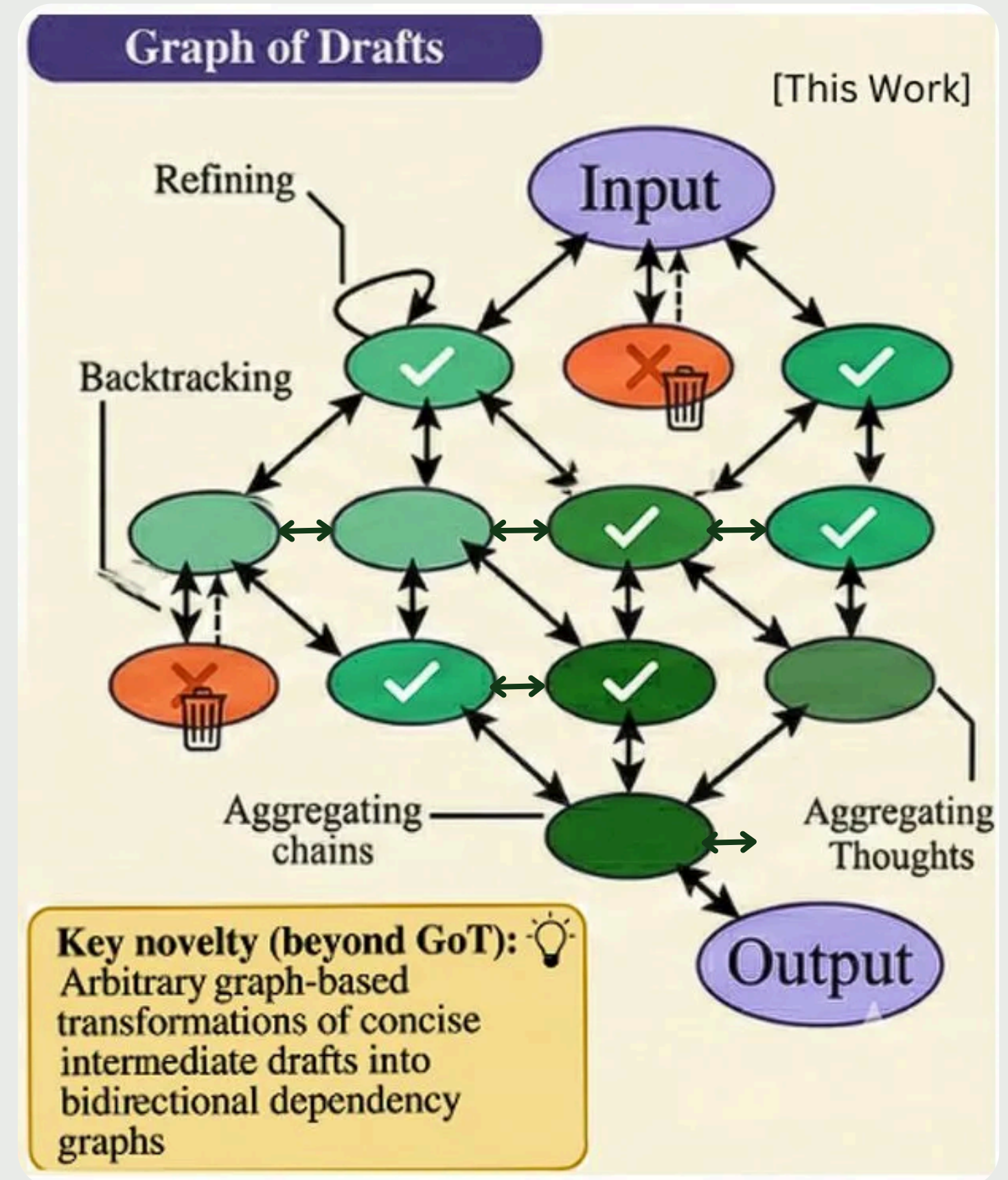
Limitations



“Every major reasoning framework for LLMs only thinks in one direction. Human cognition doesn't”

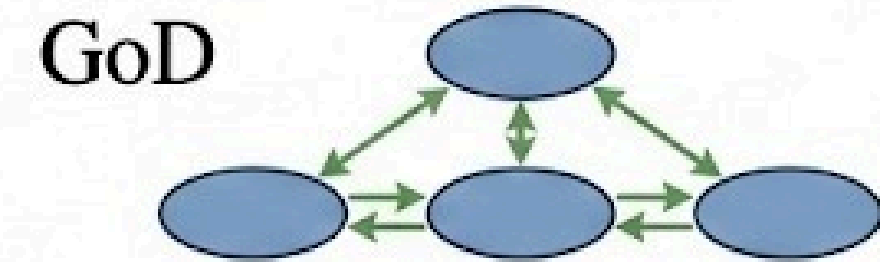
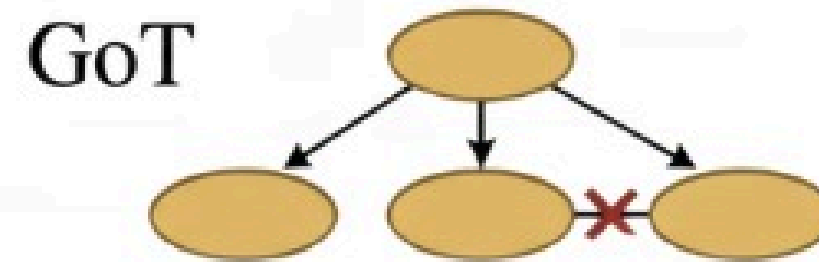
Graph of Drafts

- Graph of Drafts models LLM reasoning as a bidirected graph.
- Each node encodes a concise, information-dense draft rather than a verbose intermediate thought.
- Bidirected edges uniquely enable peer-level synergy and upstream re-contextualization, structurally absent in all prior frameworks.

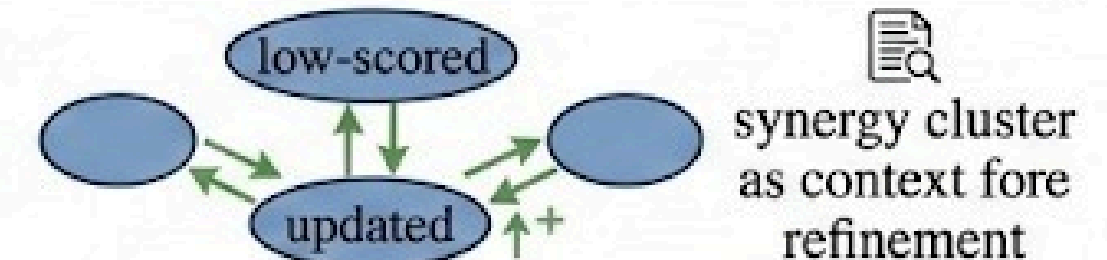
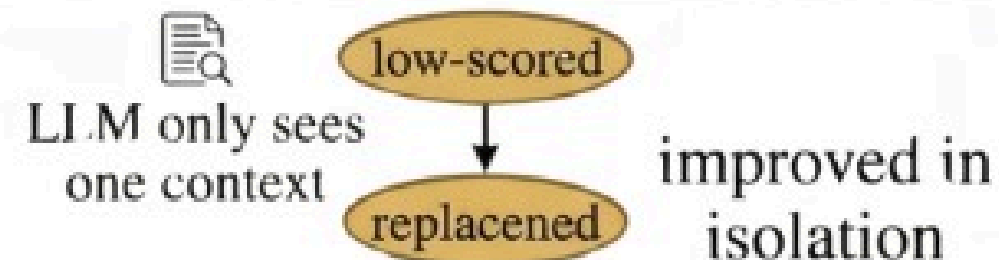


GoD over GoT: Five Structural Novelties

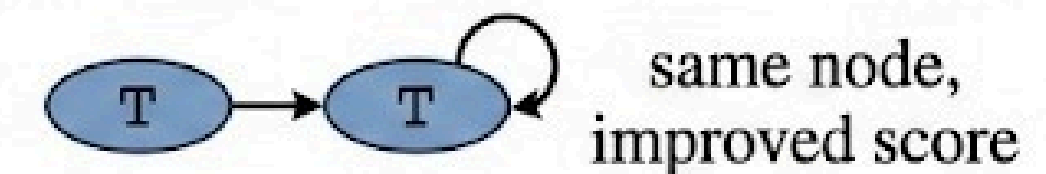
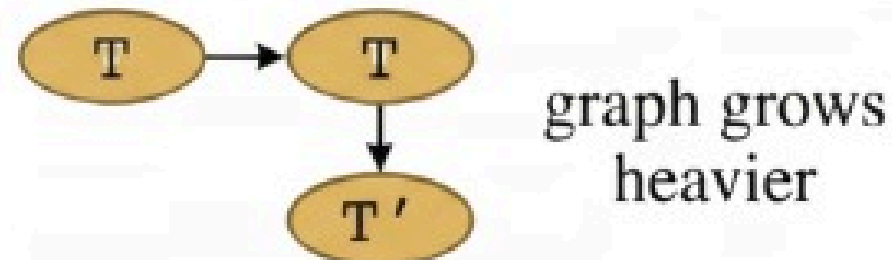
**1: Sibling Nodes
Can Communicate**



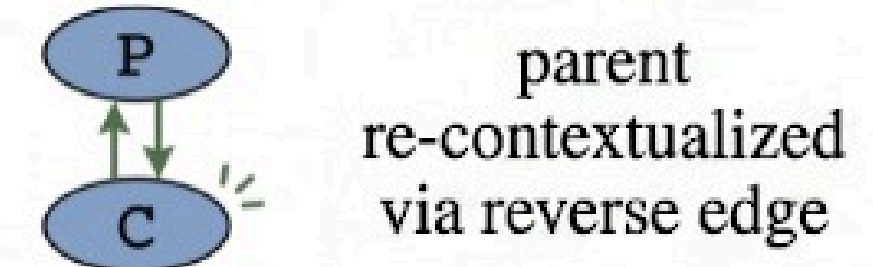
**2: Refinement
Draws on Peer Intelligence**



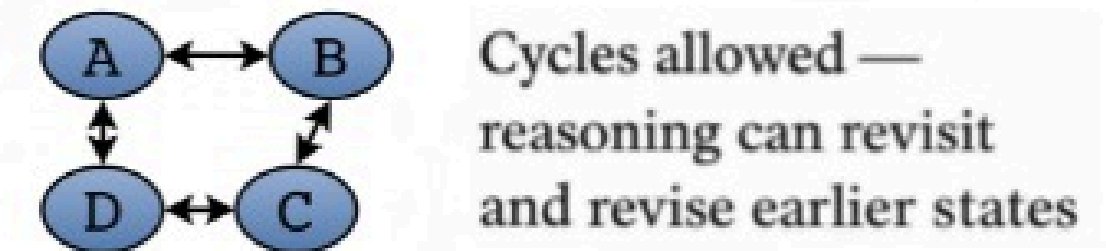
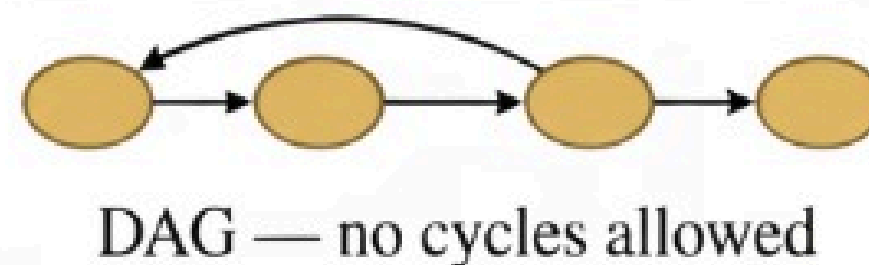
**3: Nodes Updated
In Place, Not Replaced**



**4: Refined Child
Re-Contextualizes Parent**



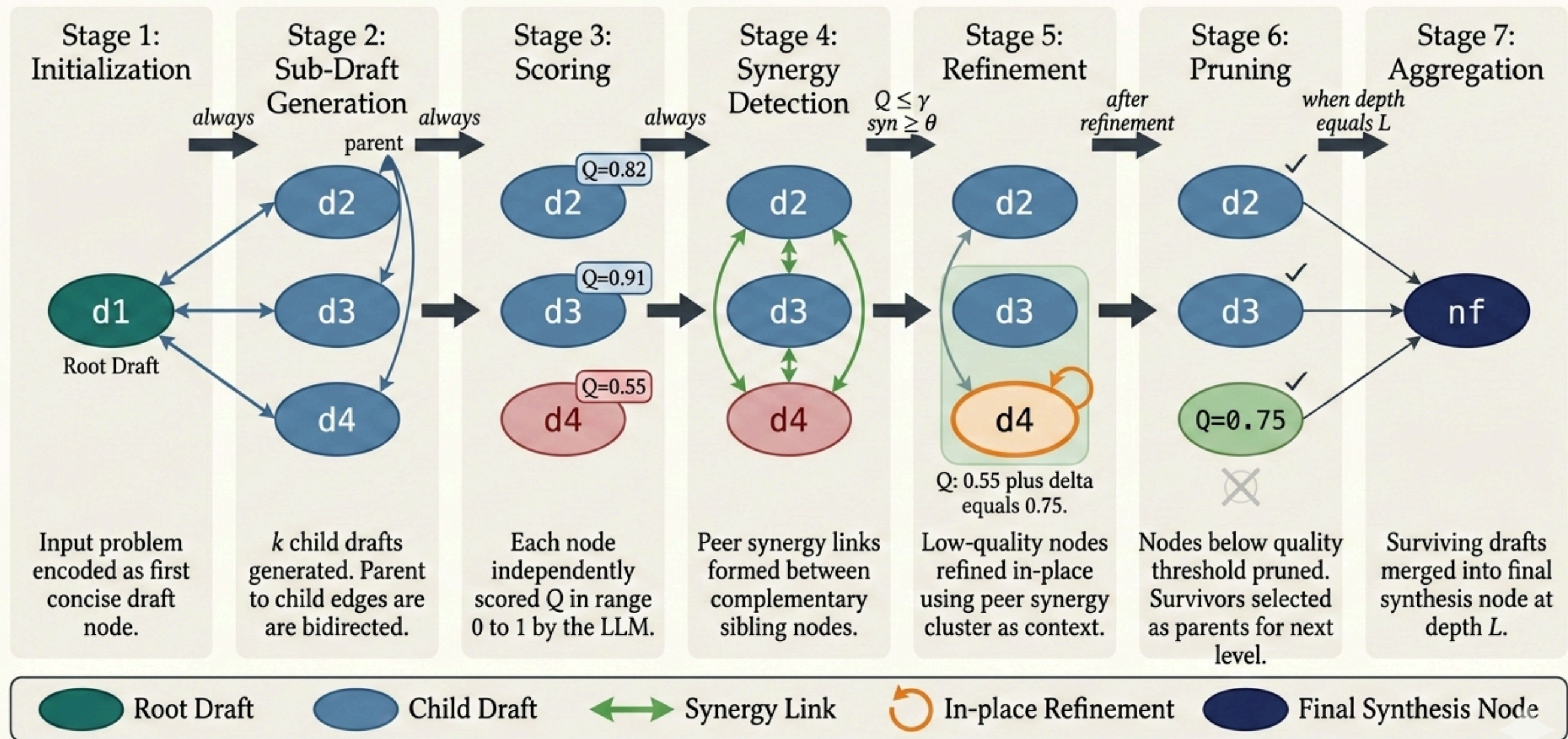
**5: Cycles Are
Structurally Permitted**



GoD's bidirected topology enables all five novelties as native structural properties, not approximations.

Operational Flow

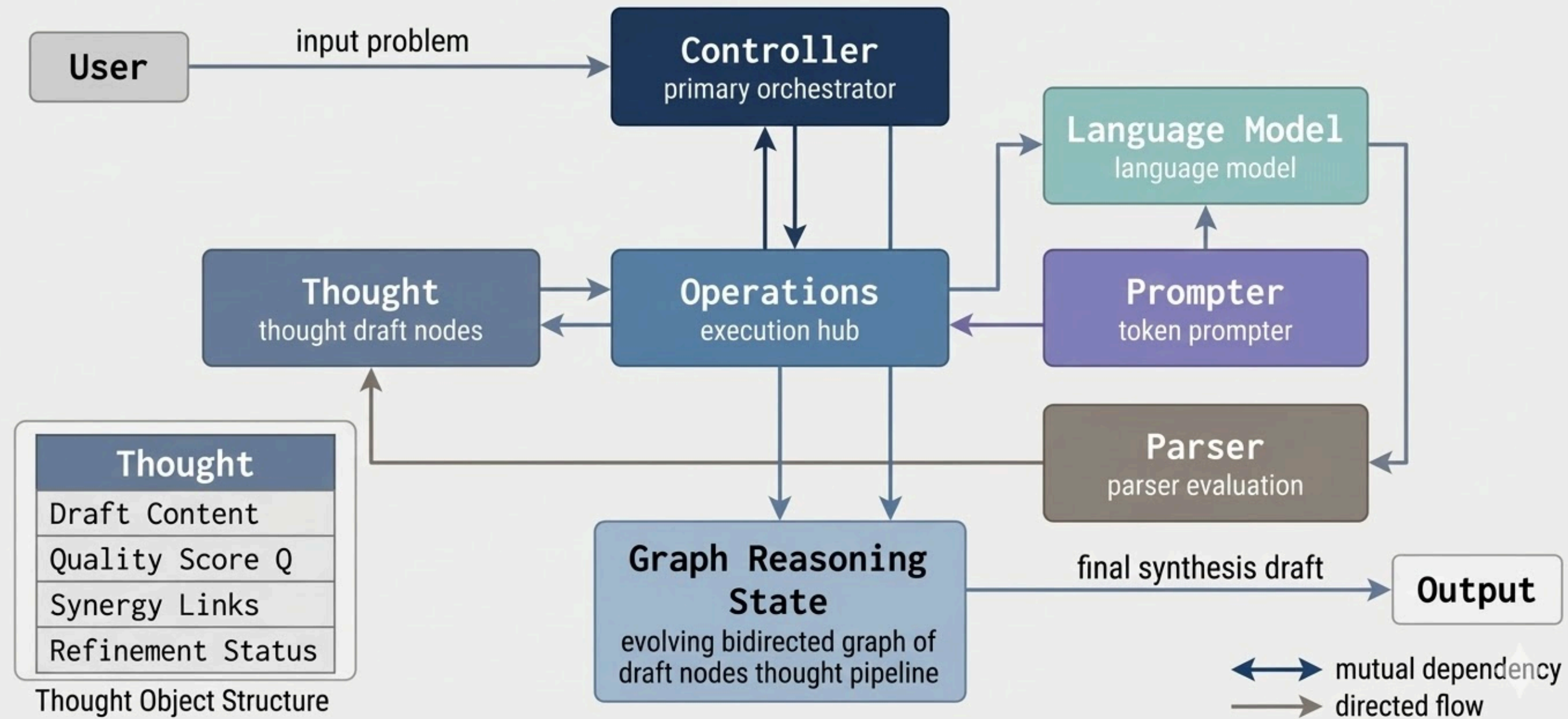
Graph of Drafts: Operational Flow



Stages 2 through 6 repeat per level until maximum depth L is reached, then aggregation produces the final answer.

System Architecture

Graph of Drafts: System Architecture



Evaluations – Set up

Parameter	Details
Model	Gemini Flash (identical across all frameworks)
Tasks	50 diverse tasks (Math, Logic, String Manipulation, Creative Synthesis)
Baselines	Chain of Drafts, Tree of Thoughts, Graph of Thoughts
Metrics	Token consumption, API calls, execution time, reasoning quality
Edge Modes	Bidirected vs. directed configurations (tested via ablation)

Evaluations – Use Cases

Name	Description
Logical Riddle	Multi-step truth table reasoning requiring strategic question formulation under logical constraints
Bayesian Inference	Posterior probability derivation from decomposed probability components and likelihood relationships
Cryptarithmic	Letter-to-digit substitution solving under carry propagation, digit uniqueness, and leading digit constraints
Route Optimisation	Optimal path generation through multiple nodes balancing minimal distance and maximum constraint satisfaction
Mathematical Pattern Recognition	Sequence completion and formula derivation through hierarchical decomposition of arithmetic and geometric relationships

Evaluations

PER-TASK BREAKDOWN

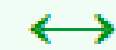
Task Type	Directed Q	Bidirected Q	Delta Q
Logical Riddle	0.74	0.82	+0.08
Cryptarithmic	0.70	0.76	+0.06
Bayesian Inference	0.66	0.71	+0.05
Route Optimisation	0.71	0.77	+0.06
Pattern Recognition	0.73	0.78	+0.05
Overall Average	0.708	0.768	+0.060

↑
+0.06

Quality Gain

Bidirected vs Directed

0.708 → 0.768 overall average



+57%

More Synergy Links

Bidirected configuration

More peer connections activated

Results & Discussions

Baseline	Baseline Avg Tokens	GoD Avg Tokens	Token Reduction
CoD	426.5	192.68	54.8%
ToT	883.7	192.68	78.2%
GoT	1294.7	192.68	85.1%

- **GoD averages 192.68 tokens per task, achieving reductions of 55% over CoD, 78% over ToT, and 85% over GoT, establishing the lowest token consumption across all evaluated frameworks**
- **Despite making 20 times more API calls than all baselines, GoD completes in 27.8 seconds against GoT's 58.7 seconds, confirming that output verbosity rather than inference call overhead is the true computational bottleneck in LLM reasoning**

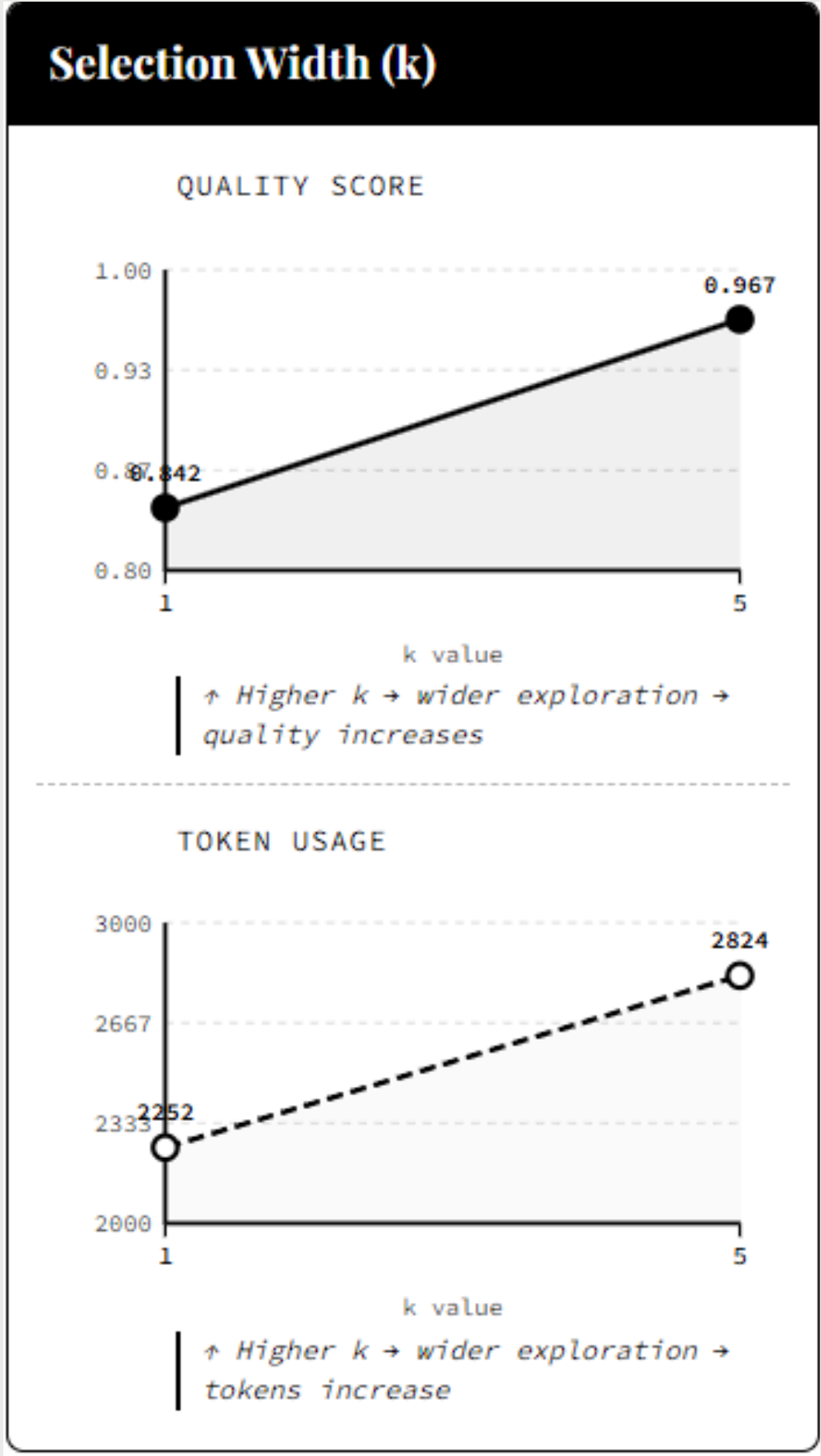
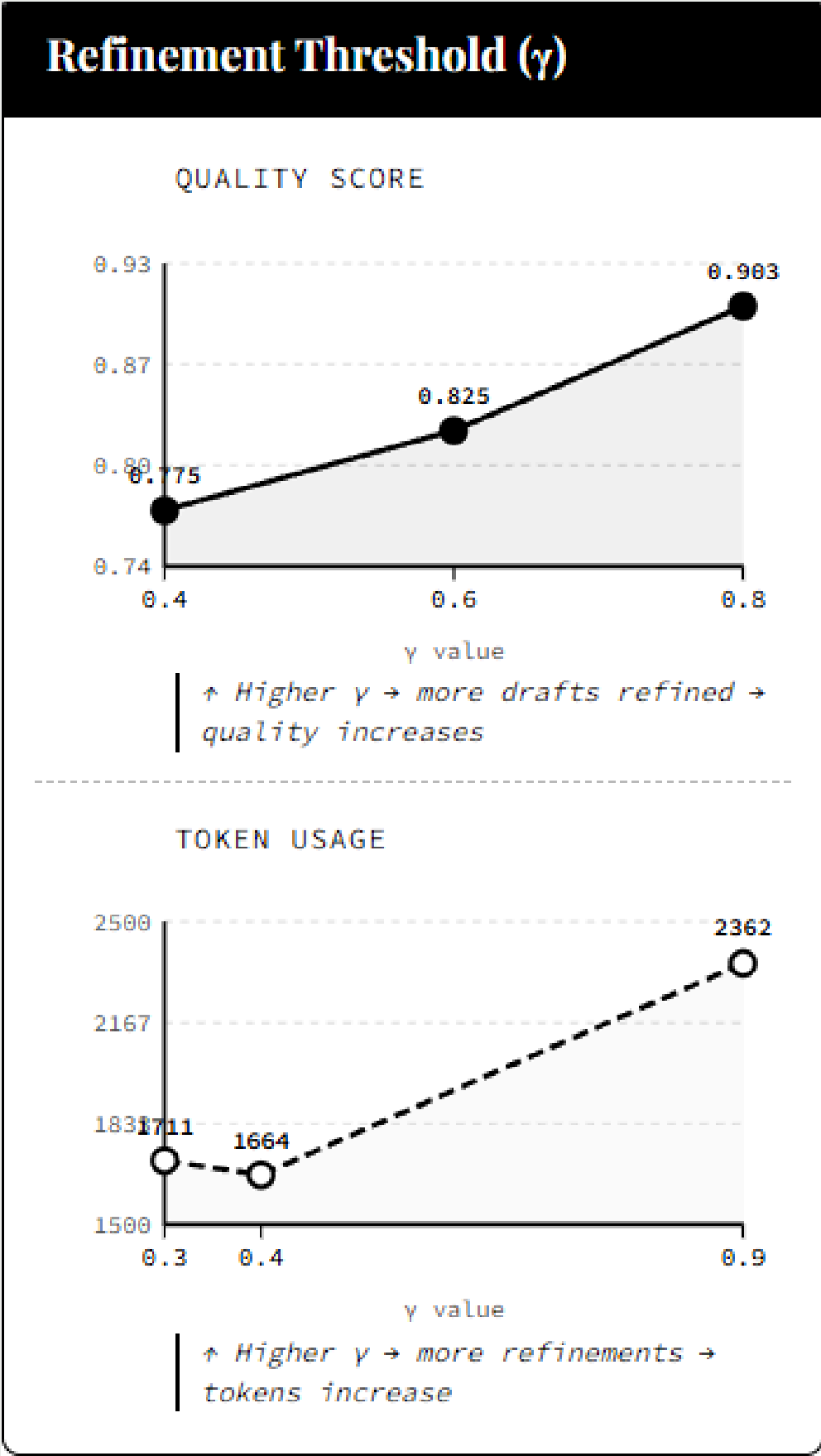
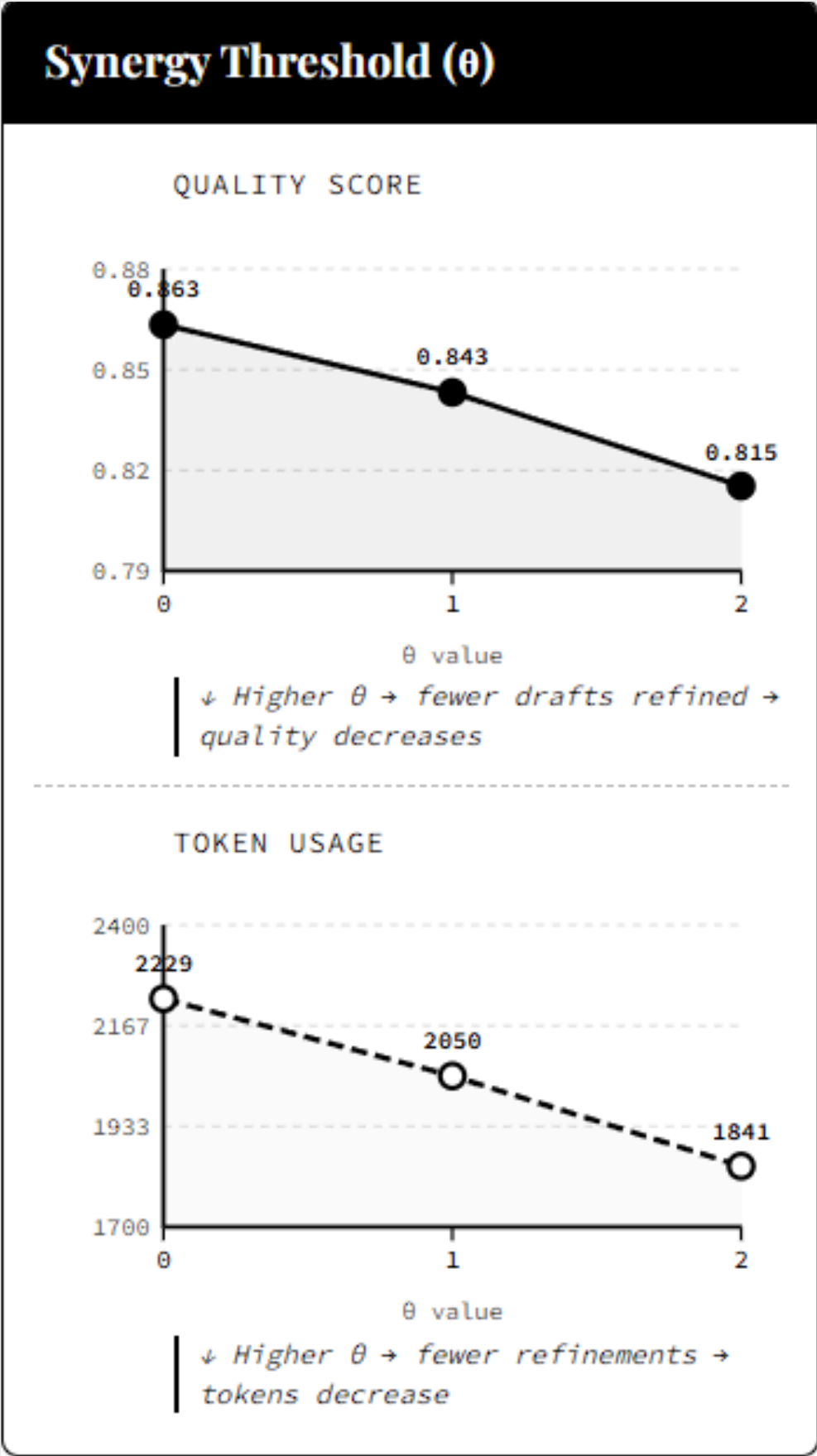
Conclusion

- GoD establishes that reasoning topology, not model scale, is the critical determinant of both reasoning quality and computational efficiency
- Bidirected edges natively enable peer-level synergy and upstream re-contextualization, two structural capabilities fundamentally absent in all prior frameworks
- GoD achieves up to 85% token reduction and 20% quality improvement over the state of the art, demonstrating that structural design outperforms output verbosity as an efficiency lever
- Future directions include extension to complex architectures, & formal specification of upstream propagation

References

No	Authors	Title	Link
1	J. Wei, X. Wang, D. Schuurmans, et al.	Chain-of-Thought Prompting Elicits Reasoning in Large Language Models	https://arxiv.org/abs/2201.11903
2	S. Yao, D. Yu, J. Zhao, et al.	Tree of Thoughts: Deliberate Problem Solving with Large Language Models	https://arxiv.org/abs/2305.10601
3	M. Besta, N. Blach, A. Kubicek, et al.	Graph of Thoughts: Solving Elaborate Problems with Large Language Models	https://arxiv.org/abs/2308.09687
4	S. Xu, W. Xie, L. Zhao, P. He	Chain of Draft: Thinking Faster by Writing Less	https://arxiv.org/abs/2502.18600
5	X. Wang, J. Wei, D. Schuurmans, et al.	Self-Consistency Improves Chain of Thought Reasoning in Language Models	https://arxiv.org/abs/2203.11171

Appendix - Sensitivity Analysis



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Thank you!

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